

Workout Tracker

A Native Android Fitness Application



Course
CSC-410 Mobile Application Development

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Abstract

Workout Tracker is a feature-rich native Android application meticulously crafted in **Kotlin** to empower fitness enthusiasts in achieving their health and performance goals. The application provides a seamless experience for logging workouts, maintaining consistent training streaks, calculating Body Mass Index (BMI) with personalized recommendations, competing on real-time public leaderboards, and securely managing comprehensive user profiles.

Built using modern Android development practices, the app leverages **Android Jetpack** components, **Material Design 3**, **Room Database** for efficient local caching, and **Firebase** for secure authentication and real-time data synchronization. With its intuitive interface, dynamic animations, and robust offline support, Workout Tracker transforms fitness tracking into an engaging and motivating journey.

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Chapter 1

Introduction

1.1 Project Overview

Workout Tracker is a comprehensive fitness companion app that motivates users through gamification and insightful analytics. The app supports workout logging, progress tracking, BMI calculation, and public leaderboards.

1.2 Objectives

- Deliver an intuitive and engaging workout logging experience
- Ensure seamless data synchronization using Firebase
- Implement secure authentication and privacy features
- Provide real-time progress insights and competitive elements

Chapter 2

Features

2.0.1 Interactive Dashboard

- Dynamic statistics (calories, streak, workouts)
- Timezone-safe streak counter
- Collapsible search bar

2.0.2 Workout Management

- Add/Edit workouts with sets, reps, weight & category
- Quick templates support
- Automatic timestamps

2.0.3 BMI Calculator

- Accurate BMI calculation with unit conversion
- Color-coded results + personalized advice
- Graceful empty state handling

2.0.4 Public Leaderboard

- Sort by Streak, Workouts, or Calories
- Top 3 medal system
- Privacy-respecting filtering

2.0.5 User Profile

- Detailed physical profile with unit support
- Profile picture stored locally (Room + Base64)
- Public/Private toggle
- Secure password change

Chapter 3

Architecture & Technology Stack

3.1 System Architecture

The application follows a clean layered architecture with a central Controller layer managing communication between the View and various data sources.

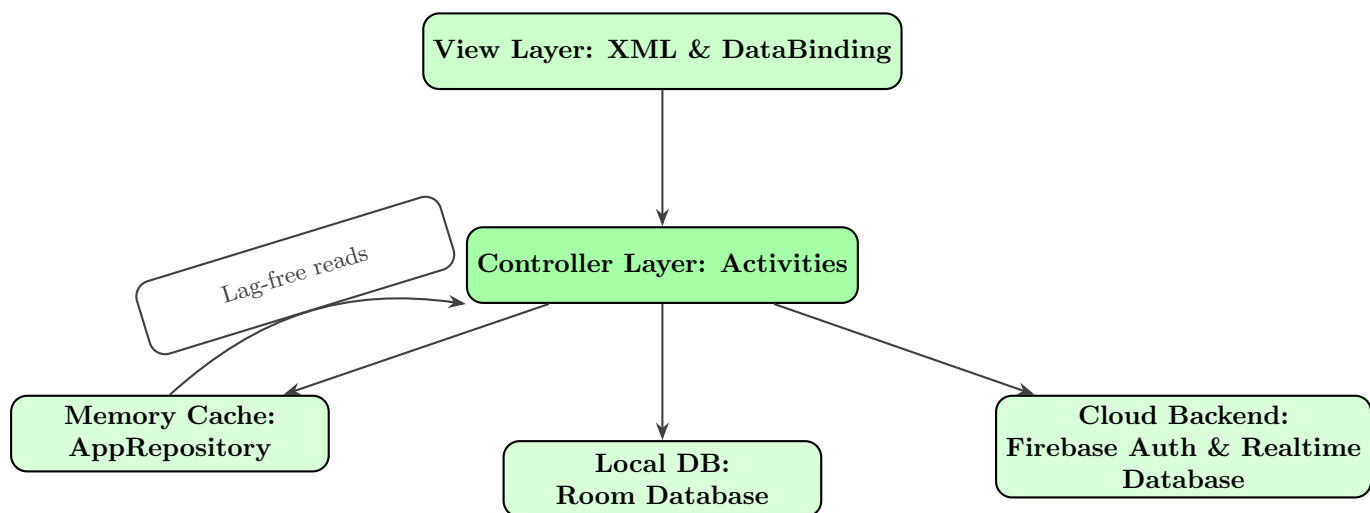


Figure 3.1: Application Architecture Diagram

This diagram clearly shows:

- **Presentation Layer:** User Interface (XML + Material Design)
- **Business Logic Layer:** Activities/Fragments + Repository
- **Data Layer:** Local persistence (Room) and Cloud backend (Firebase)

3.2 Technology Stack

Component	Technology
Programming Language	Kotlin
User Interface	XML Layouts + Material Design 3
Architecture Pattern	MVVM + Repository Pattern
Local Database	Room (SQLite)
Cloud Backend	Firebase Authentication & Realtime Database
Image Handling	Base64 encoding stored in Room
Caching	In-memory Repository + Room
Build Tool	Gradle
Minimum SDK	API 24
Target SDK	API 34+

Table 3.1: Tech Stack

Chapter 4

Screenshots

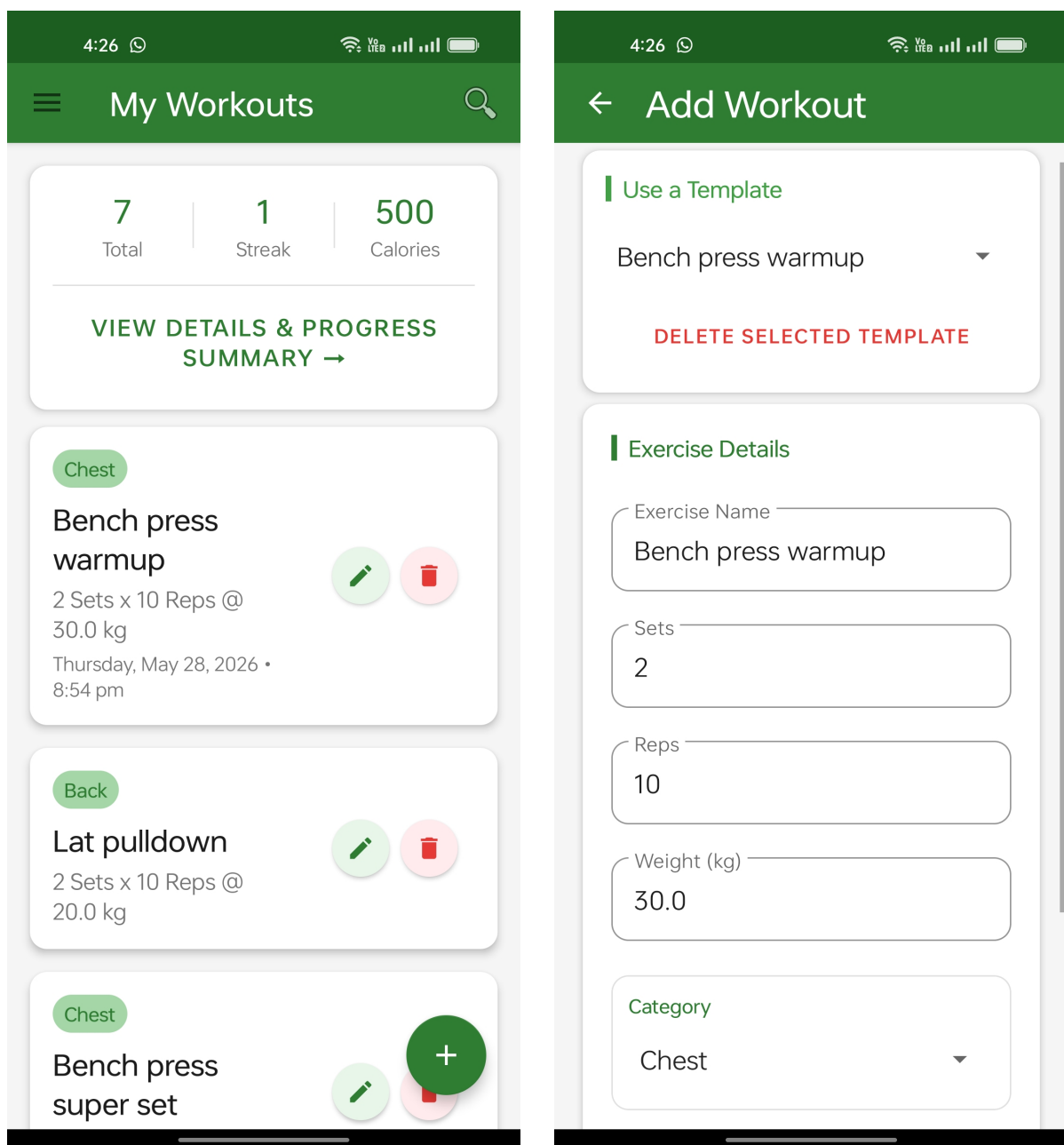


Figure 4.1: Dashboard and Add Workout Screen

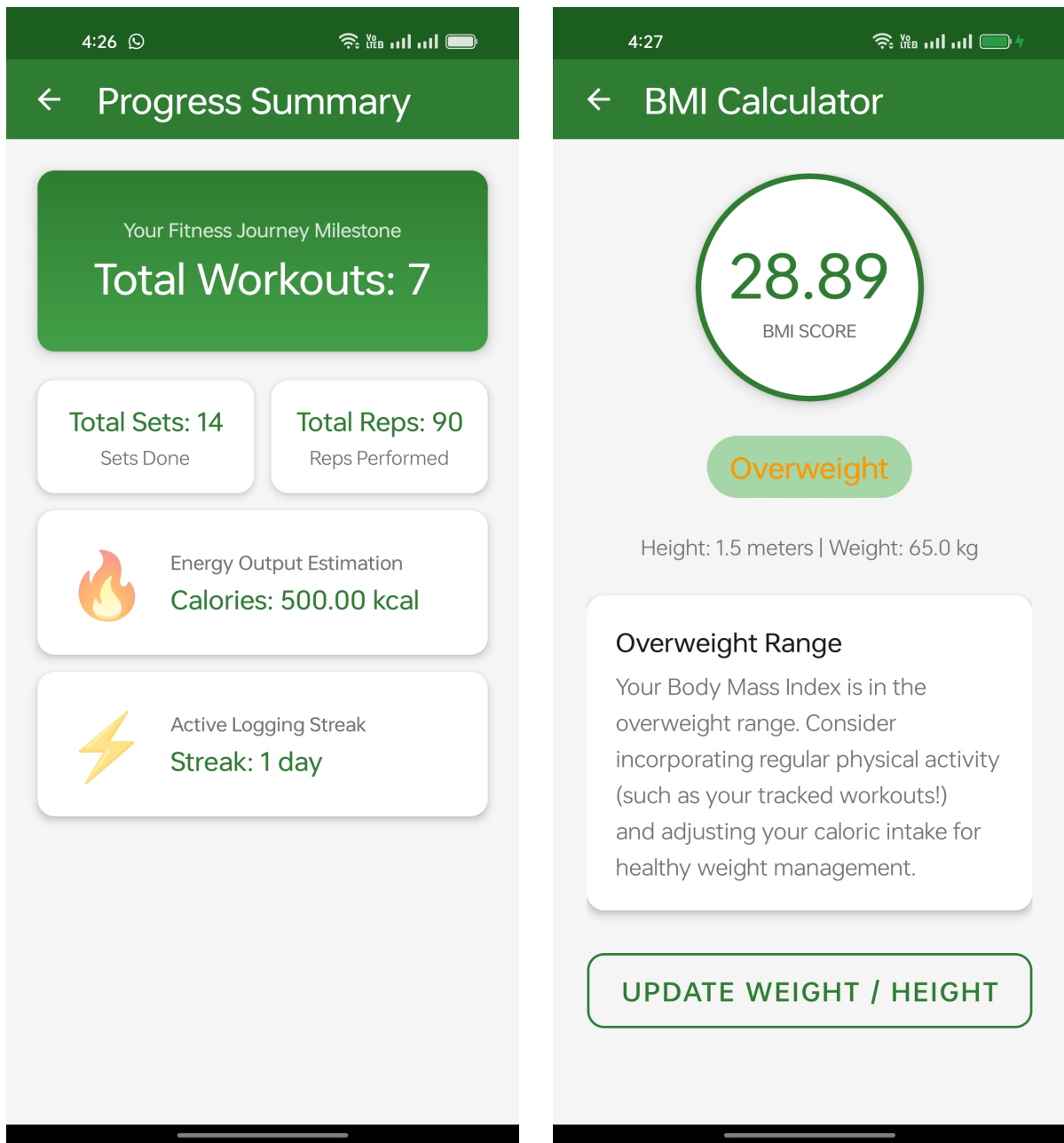


Figure 4.2: Progress Summary and BMI Calculator

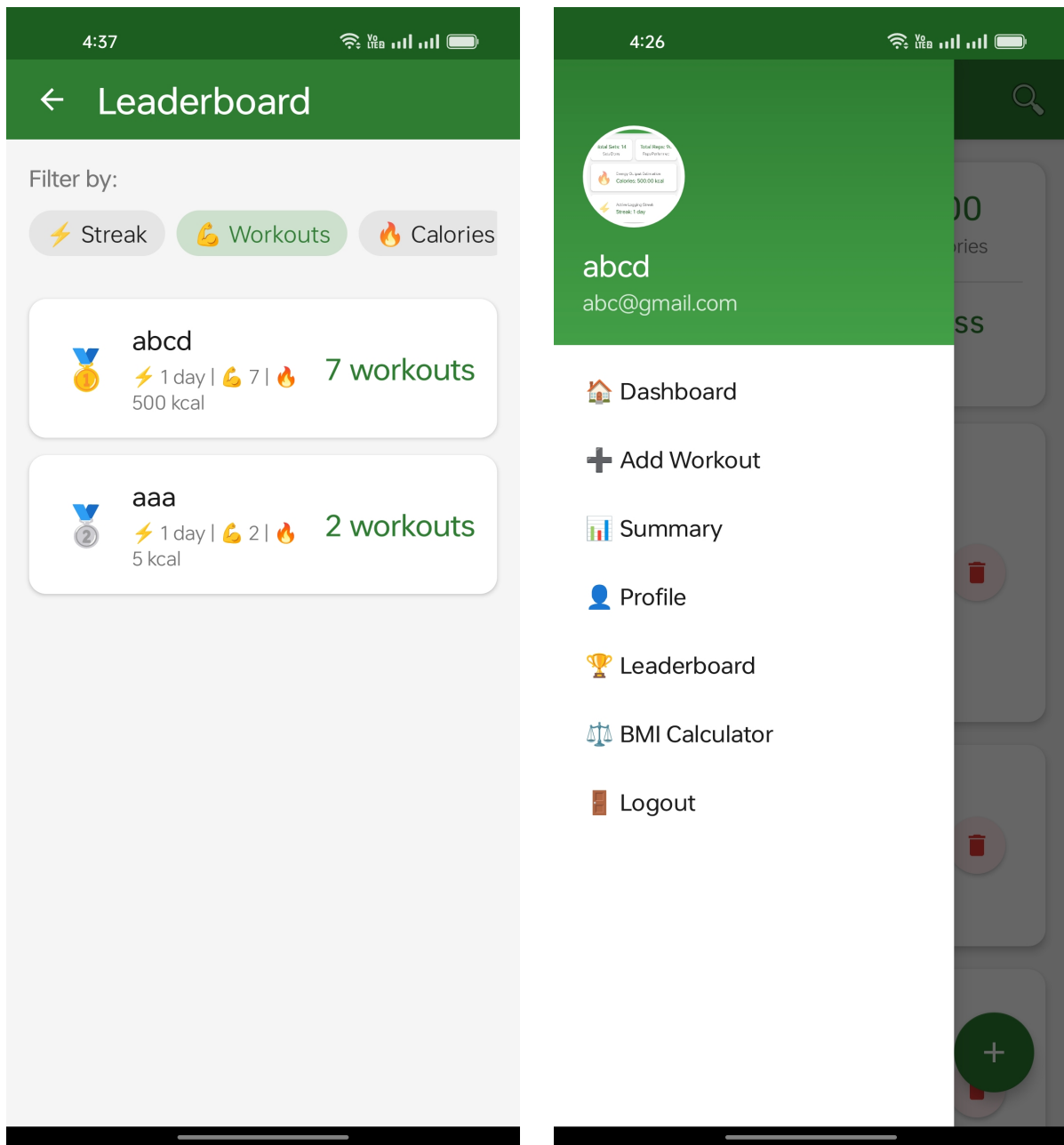


Figure 4.3: Leaderboard and Navigation Drawer

4:53

Vg LTEB

←

My Profile

Age

23

Gender

☒ Male

☐ Female

Weight

65.0

Unit: kg

Height

1.5

Unit: meters

Profile Privacy

Allow users to see your workout stats.

Public Profile

☒

SAVE CHANGES

CHANGE PASSWORD

Figure 4.4: User Profile Screen

Chapter 5

Setup & Installation

5.1 Installation Steps

5.1.1 Prerequisites

- Android Studio (Koala or later)
- JDK 17 or higher
- Firebase Project with `google-services.json`

5.1.2 Steps

1. Clone the repository: `git clone https://github.com/MianSaadTahir/workout-tracker`
2. Place `google-services.json` in `app/` folder
3. Sync project with Gradle
4. Build & Run (`./gradlew assembleDebug`)

Chapter 6

Conclusion & Future Work

The Workout Tracker successfully demonstrates modern Android development practices with a focus on user experience, data security, and performance.

Future enhancements may include:

- Workout exercise library with animations
- Social features (following friends)
- Advanced analytics and charts
- Wear OS integration
- Dark mode improvements